Because the given polynomial is a perfect square $a-16=0$, $b-16=0$
Therefore, $a=16$, $b=16$.



**Exercise 3.8**

1. Find the square root of the following polynomials by division method

   (i) $x^4-12x^3+42x^2-36x+9$ (ii) $37x^2-28x^3+4x^4+42x+9$

   (iii) $16x^4+8x^2+1$ (iv) $121x^4-198x^3-183x^2+216x+144$

2. Find the values of $a$ and $b$ if the following polynomials are perfect squares

   (i) $4x^4-12x^3+37x^2+bx+a$ (ii) $ax^4+bx^3+361x^2+220x+100$

3. Find the values of $m$ and $n$ if the following polynomials are perfect sqaures

   (i) $36x^4-60x^3+61x^2-mx+n$ (ii) $x^4-8x^3+mx^2+nx+16$

## 3.6 Quadratic Equations

### Introduction

Arab mathematician Abraham bar Hiyya Ha-Nasi, often known by the Latin name Savasorda, is famed for his book 'Liber Embadorum' published in 1145 AD(CE) which is the first book published in Europe to give the complete solution of a quadratic equation.

For a period of more than three thousand years beginning from early civilizations to current times, humanity knew how to solve a general quadratic equation in terms of its co-efficients by using four arithmetical operations and extraction of roots. This process is called "Solving by Radicals". Huge amount of research has been carried to this day in solving various types of equations.

### Quadratic Expression

An expression of degree $n$ in variable $x$ is $a_0x^n+a_1x^{n-1}+a_2x^{n-2}+\ldots+a_{n-1}x+a_n$ where $a_0\neq 0$ and $a_1, a_2, \ldots a_n$ are real numbers. $a_0, a_1, a_2, \ldots a_n$ are called coefficients of the expression.

In particular an expression of degree 2 is called a Quadratic Expression which is expressed as $p(x)=ax^2+bx+c$, $a\neq 0$ and $a, b, c$ are real numbers.

### 3.6.1 Zeroes of a Quadratic Polynomial

Let $p(x)$ be a polynomial. $x=a$ is called zero of $p(x)$ if $p(a)=0$

For example, if $p(x)=x^2-2x-8$ then $p(-2)=4+4-8=0$ and $p(4)=16-8-8=0$

Therefore, $-2$ and $4$ are zeros of the polynomial $p(x)=x^2-2x-8$.



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